

Chapter 4 Higher order Differential Equations

In this chapter, we look at DEs of order two or higher. We are only considering Linear equations. Most of our examples will be for equations of order two but the general theory of Linear DEs will be presented for n^{th} order equations.

4.1 Preliminary theory - Linear Equations

Consider the n^{th} order Linear DE:

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = g(x).$$

An IVP would have n initial conditions.

$$y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1}$$

If we assume that :

$$a_n(x), \dots, a_1(x), a_0(x), g(x)$$

are continuous and $a_n(x) \neq 0$ for all x in an interval I containing x_0 ,

then the IVP :

$$\begin{cases} a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = g(x) \\ y(x_0) = y_0, y'(x_0) = y_1, \dots, y^{(n-1)}(x_0) = y_{n-1} \end{cases}$$

has a unique solution $y(x)$ over interval I .

Homogeneous Equations

An equation of the form :

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = 0 \quad [H]$$

is said to be homogeneous.

The zero makes
it homogeneous.

Solutions of the homogeneous equation [H]
satisfy the Superposition Principle.

If $y_1, y_2, \dots, y_k$ are solutions of [H],

then :

$$c_1 y_1 + c_2 y_2 + \dots + c_k y_k$$

Linear combination

is also a solution of [H] for any
constants $c_1, c_2, \dots, c_k$.

ex: ① $y_1 = \cos x$ and $y_2 = \sin x$

are solutions of :

$$y'' + y = 0. \quad [H]$$

By the superposition Principle,

$$y = \underbrace{C_1 y_1 + C_2 y_2}$$

Linear Combination of y_1, y_2

$$y = C_1 \cos x + C_2 \sin x$$

is also a solution of $[H]$

for any constants C_1, C_2 .

Note : The superposition principle applies because $[H]$ is Linear and homogeneous.

Recall from Linear Algebra

If S is a subspace of dimension n , then a basis of S consists of

n vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ such that:

① $\text{span}(\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n) = S$

② The vectors $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ are linearly independent.

Any vector $\vec{v}$ in S can be expressed uniquely as a linear combination of the basis vectors.

$$\vec{v} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_n \vec{v}_n$$

By analogy with Linear Algebra,

the solutions of :

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = 0 \quad [H]$$

form a Subspace of dimension n .

The functions that are solutions of $[H]$ play the role of vectors.

Solving $[H]$ requires that we find a "Basis" for the solutions of $[H]$.

Such a basis consists of n Linearly independent solutions of $[H]$.

We will now provide precise definitions for all of this.

Definition A set of functions

$y_1(x), y_2(x), \dots, y_n(x)$ defined
on an interval I is Linearly dependent

if there are constants $c_1, c_2, \dots, c_n$
not all zero, so that :

$$c_1 y_1(x) + c_2 y_2(x) + \dots + c_n y_n(x) = 0$$

for all x in I .

Otherwise, they are Linearly independent.

Notation

LD $\rightarrow$ Linearly dependent

LI $\rightarrow$ Linearly independent.

ex: (2) Consider the functions

$$y_1 = x, \quad y_2 = x^2, \quad y_3 = 6x - 5x^2.$$

Are they LD or LI ?

Answer: They are LD since

$$-6y_1 + 5y_2 + y_3 = 0$$

$$-6x + 5x^2 + (6x - 5x^2) = 0$$

for all x .

Usually we are interested in LI functions. To show that a set of functions is LI we can use a Wronskian.

Definition

Consider the functions $y_1, y_2, \dots, y_n$.

Assume that each of them possesses at least $n-1$ derivatives.

The $n \times n$ determinant :

$$W = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

is called the Wronskian of the functions.

The Wronskian of $y_1, y_2, \dots, y_n$ can be denoted as :

$$W = W(y_1, y_2, \dots, y_n).$$

Theorem

If $y_1, y_2, \dots, y_n$ are solutions of:

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = 0$$

over an interval I , then:

$y_1, y_2, \dots, y_n$ are L I

if and only if

$$W = W(y_1, y_2, \dots, y_n) \neq 0$$

for all x in I .

ex: ③ $y_1 = \cos x$ and $y_2 = \sin x$

are solutions of $y'' + y = 0$.

Are they L I ?

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W = \cos^2 x + \sin^2 x = 1$$

$$W \neq 0 \text{ for all } x$$

$\Rightarrow \cos x$ and $\sin x$ are L I.

Consider the Homogeneous DE

$$a_n(x) y^{(n)} + \dots + a_1(x) y' + a_0(x) y = 0. [H]$$

Definition A basis [or fundamental set of solutions] for the solutions of [H] consists of n LI solutions.

If $y_1, y_2, \dots, y_n$ is a basis for the solutions of [H], then any solution of [H] can be expressed uniquely as :

$$y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n .$$

We call this $y = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$ the general solution of [H].

Nonhomogeneous Equations

An equation of the form

$$a_n(x)y^{(n)} + \dots + a_1(x)y' + a_0(x)y = g(x) \quad [NH]$$

is said to be nonhomogeneous if $g(x)$ is not identically zero.

Its associated homogeneous equation is

$$a_n(x)y^{(n)} + \dots + a_1(x)y' + a_0(x)y = 0. \quad [H]$$

Any function y_p that satisfies $[NH]$ is called a particular solution of the equation.

Theorem

Let y_p be any particular solution of $[NH]$.

If $y_1, y_2, \dots, y_n$ are LI solutions of $[H]$, then any solution of $[NH]$ can be expressed uniquely as:

$$y = \underbrace{c_1 y_1 + c_2 y_2 + \dots + c_n y_n}_{y_c} + y_p$$

for some constants $c_1, c_2, \dots, c_n$.

Note: $y_c = c_1 y_1 + c_2 y_2 + \dots + c_n y_n$

is called the complementary solution
(or the homogeneous solution).

$$y = y_c + y_p$$

is called the general solution
of $[NH]$.

ex: ④ Consider the DE

$$y'' + y = \sin x \quad [NH].$$

In example ③ we saw that

$$y_1 = \cos x \text{ and } y_2 = \sin x$$

are LI solutions of

$$y'' + y = 0. \quad [H]$$

Then: $y_c = c_1 \cos x + c_2 \sin x.$

A particular solution of $[NH]$

is $y_p = -\frac{1}{2} x \cos x$.

You can check it by plugging y_p in $[NH]$.

Therefore, the general solution of $[NH]$ is :

$$y = y_c + y_p$$

$$y = c_1 \cos x + c_2 \sin x - \frac{1}{2} x \cos x.$$

In the rest of chapter 4, we will learn techniques to find both y_c and y_p by starting from a DE.